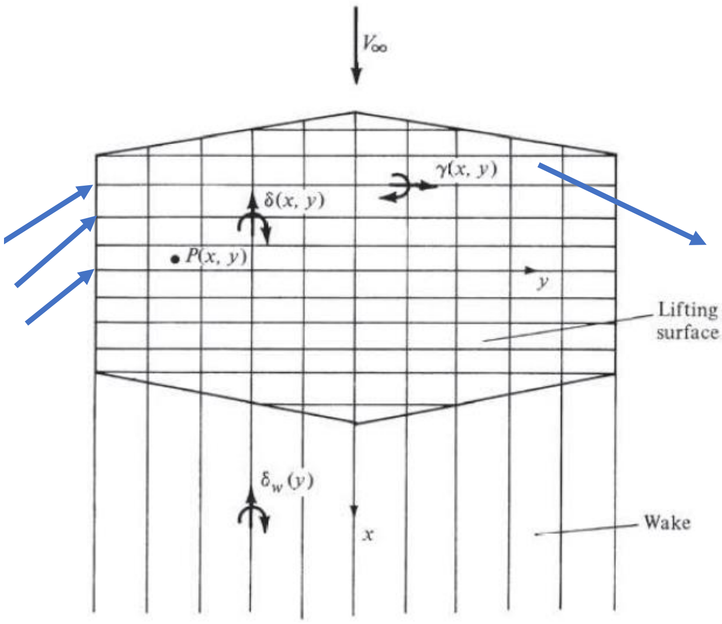
Lifting Surface Theory

- Prandtl's Lifting Line Theory gives reasonable results for straight wings at moderate to high aspect ratio (AR).
- For low AR straight wings (1 < AR < 5), Swept wings, and Delta wings, classical LLT is inappropriate.
- A simple lifting line spans the wing, with its associated trailing vortices.
- The circulation Γ varies with *y* along the lifting line. Let us extend this model by placing a *series* of lifting lines on the plane of the wing, at different chordwise stations; that is, consider a large number of lifting lines all parallel to the *y-axis*, located at different values of *x*,

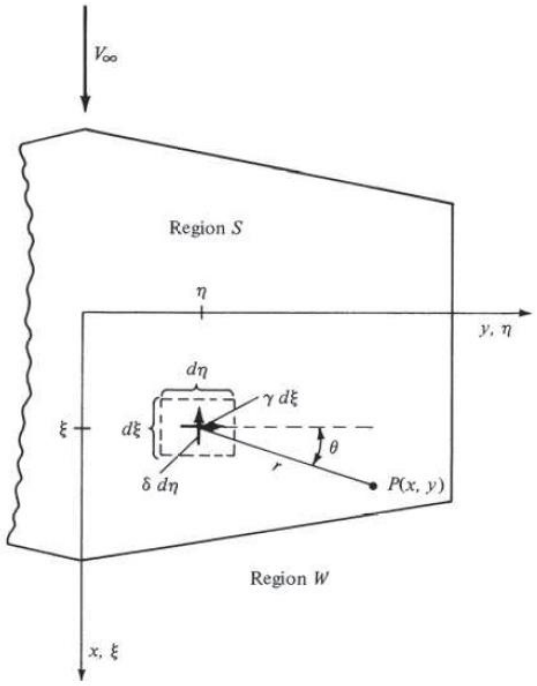
Lifting lines placed at different chordwise stations parallel to the y-axis.



Lifting surface

Vortex sheets are formed by the lifting line

- $\boldsymbol{\gamma}$ – strength of the sheet per unit length in the x-direction.
- γ – varies in the y-direction.
- $\gamma(x, y)$ – Spanwise vortex strength distribution.
- $\delta(x, y)$ – Chordwise vortex strength distribution.
- $\delta_w(x, y)$ – Strength at the trailing edge of the wing.
- Velocity induced at point *P* by an infinitesimal segment of the lifting surface and the wake. The velocity is perpendicular to the plane of the paper.



- $w(x,y) = A + B$
- $A = \frac{1}{4\pi}\iint_S \frac{(x-\xi)\gamma(\xi,\eta) + (y-\eta)\delta(\xi,\eta)}{\left[(x-\xi)^2 + (y-\eta)^2\right]^{3/2}} d\xi\, d\eta$
- $B = -\frac{1}{4\pi}\iint_W \frac{(y-\eta)\delta_w(\eta)}{\left[(x-\xi)^2 + (y-\eta)^2\right]^{3/2}} d\xi\, d\eta$
- Boundary condition:

$$w(x,y) = 0$$